

Humanoid Academy

Tag 1: Einführung und
Entwicklungsumgebung



Zeitplan

Tag 1

Mo 15.06.2026

9:30 - Open Registrierung

10:00 – 12:00 Schulungsblock 1

12:00 – 13:00 Mittagessen

13:00 – 14:30 Schulungsblock 2

14:30 – 15:00 Pause

15:00– 17:00 Schulungsblock 3



Tagesziele

- Überblick G1 Base/EDU
- Safety
- Remote Steuerung
- G1 Programmierung
- Verbindungsmethoden
- Entwicklungsumgebung



EF Robotics

- Integrator
- Distributor AMR, Reinigungsroboter, Humanoide
- Über 450 Roboter Installationen
- Eigenes Servicetechnikernetz

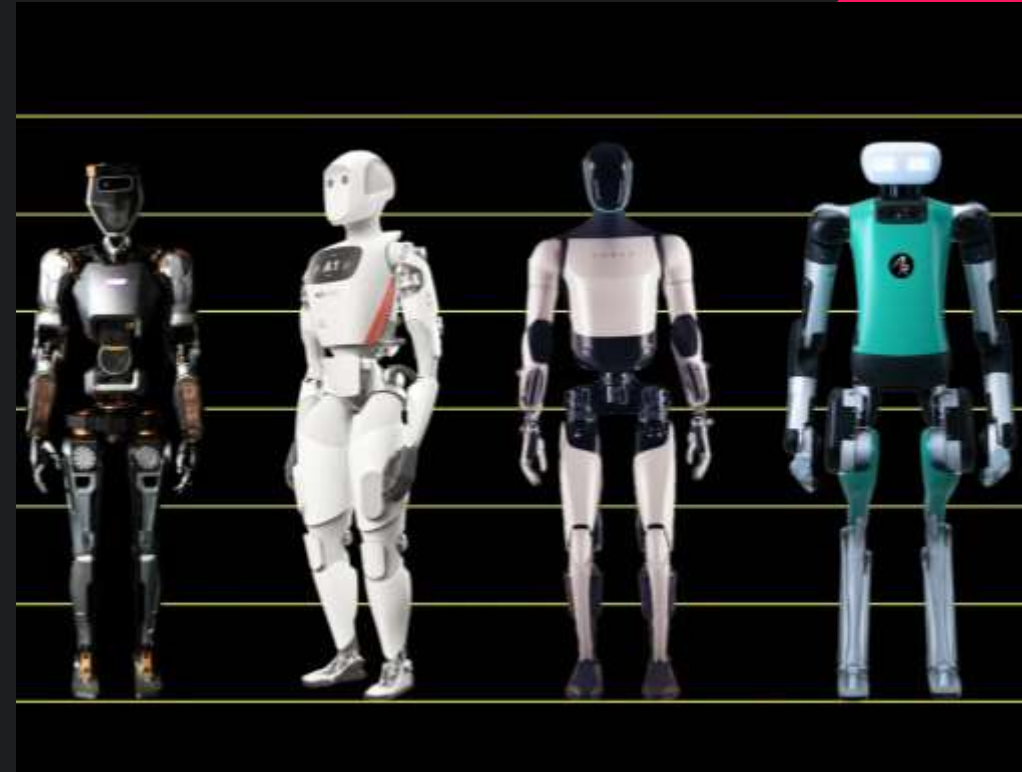


www.ef-robotics.de



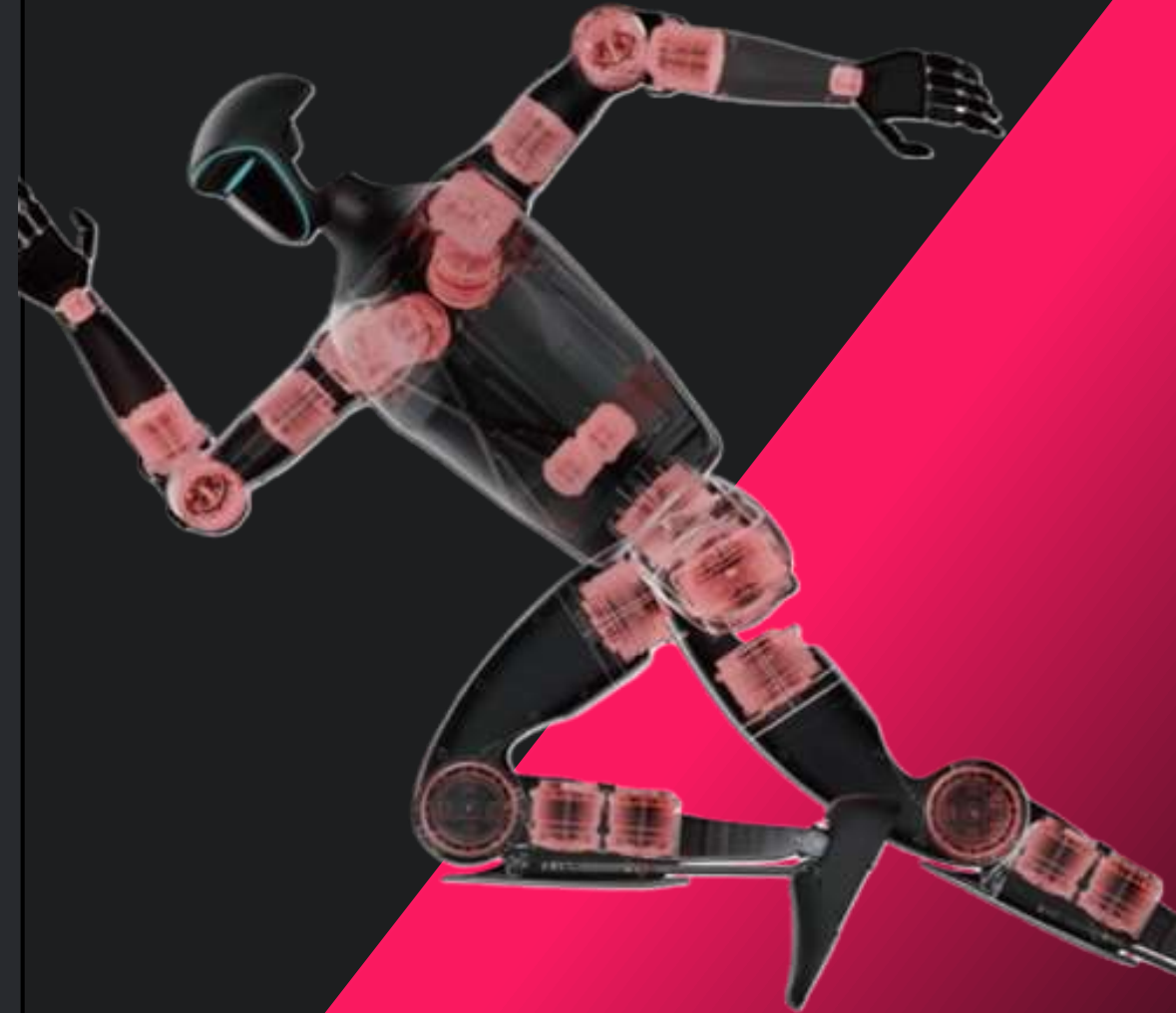
Warum Humanoide

- Universell einsetzbar
- Skalierbar
- EDU vs. Echteinsatz



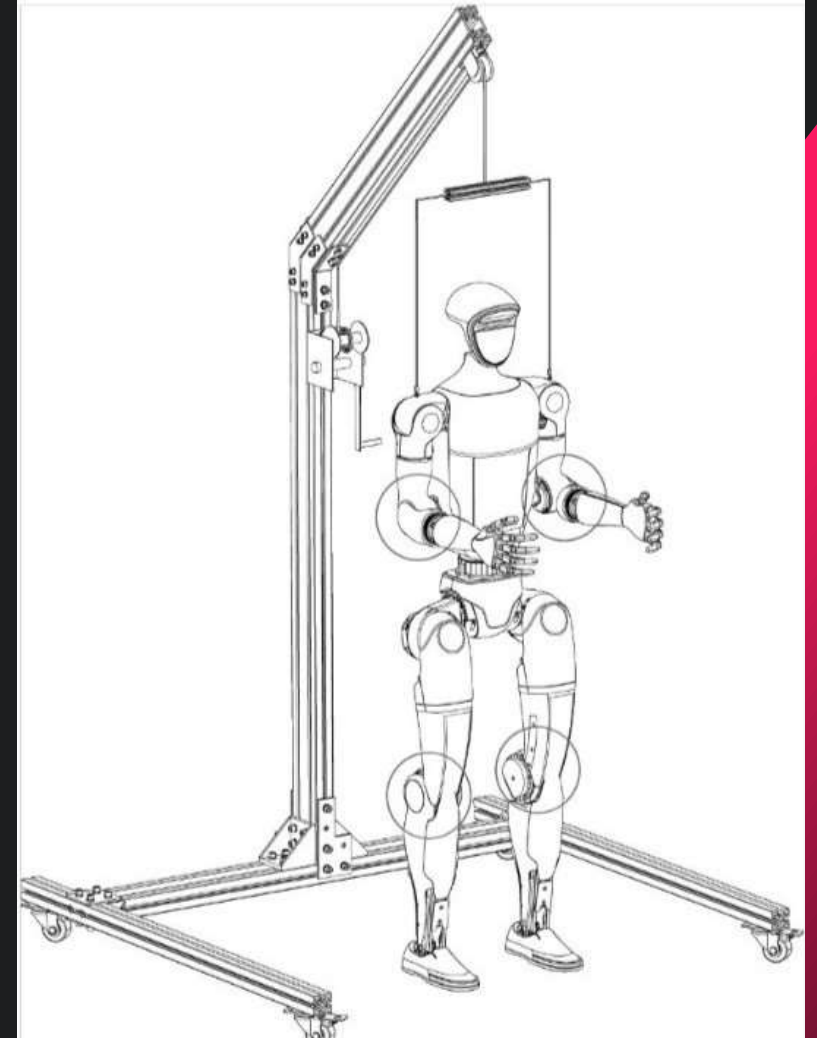
Überblick Unitree G1

Allgemein	
Gewicht	35 kg
Größe	1270 mm
Anzahl Freiheitsgrade	Up to 43
Maximaler Gelenk-Drehmoment	120 N.m

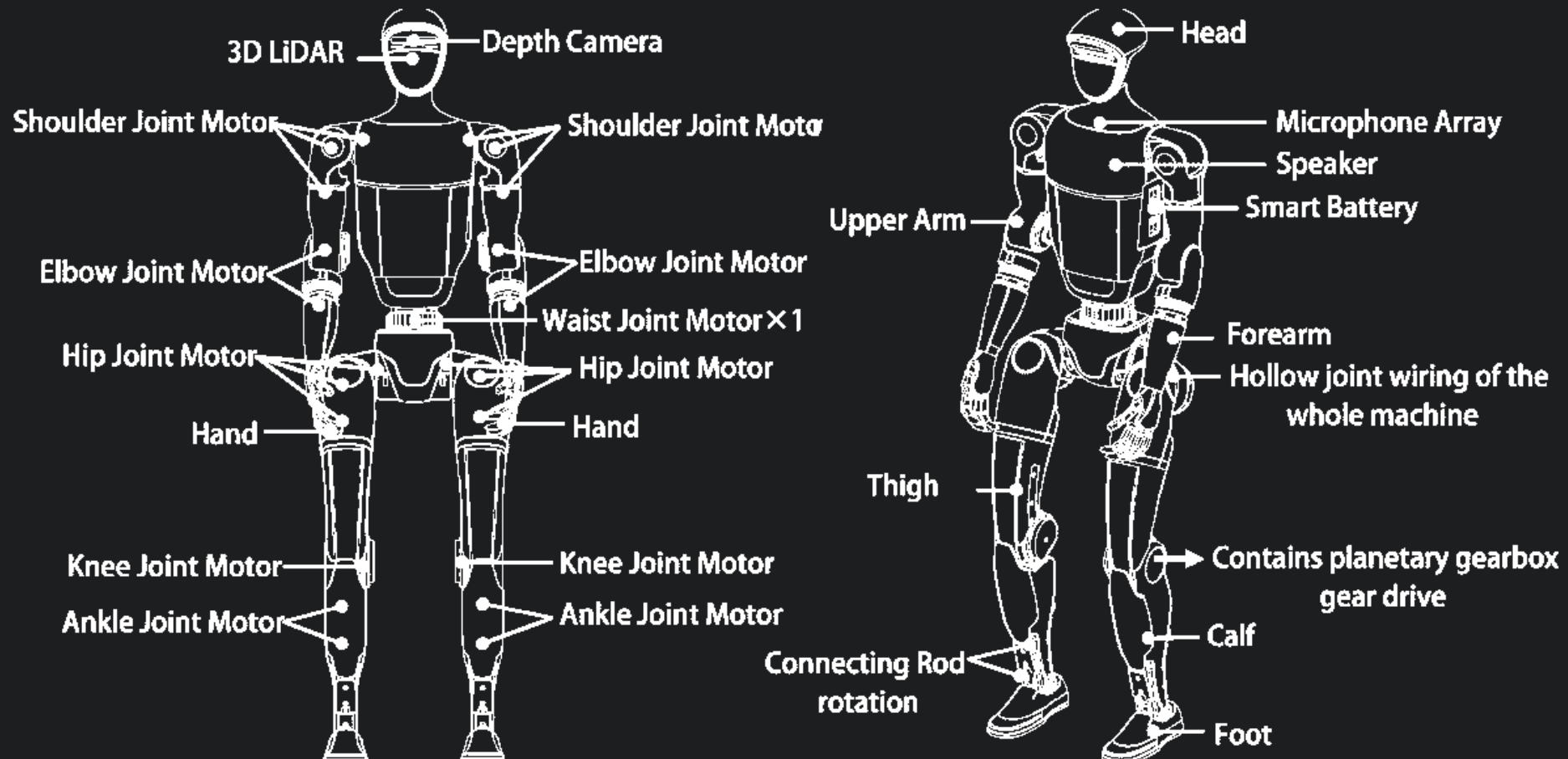


Safety

- Sicherheitszonen
- Not-Aus (Damping)
- Laborbetrieb / CE

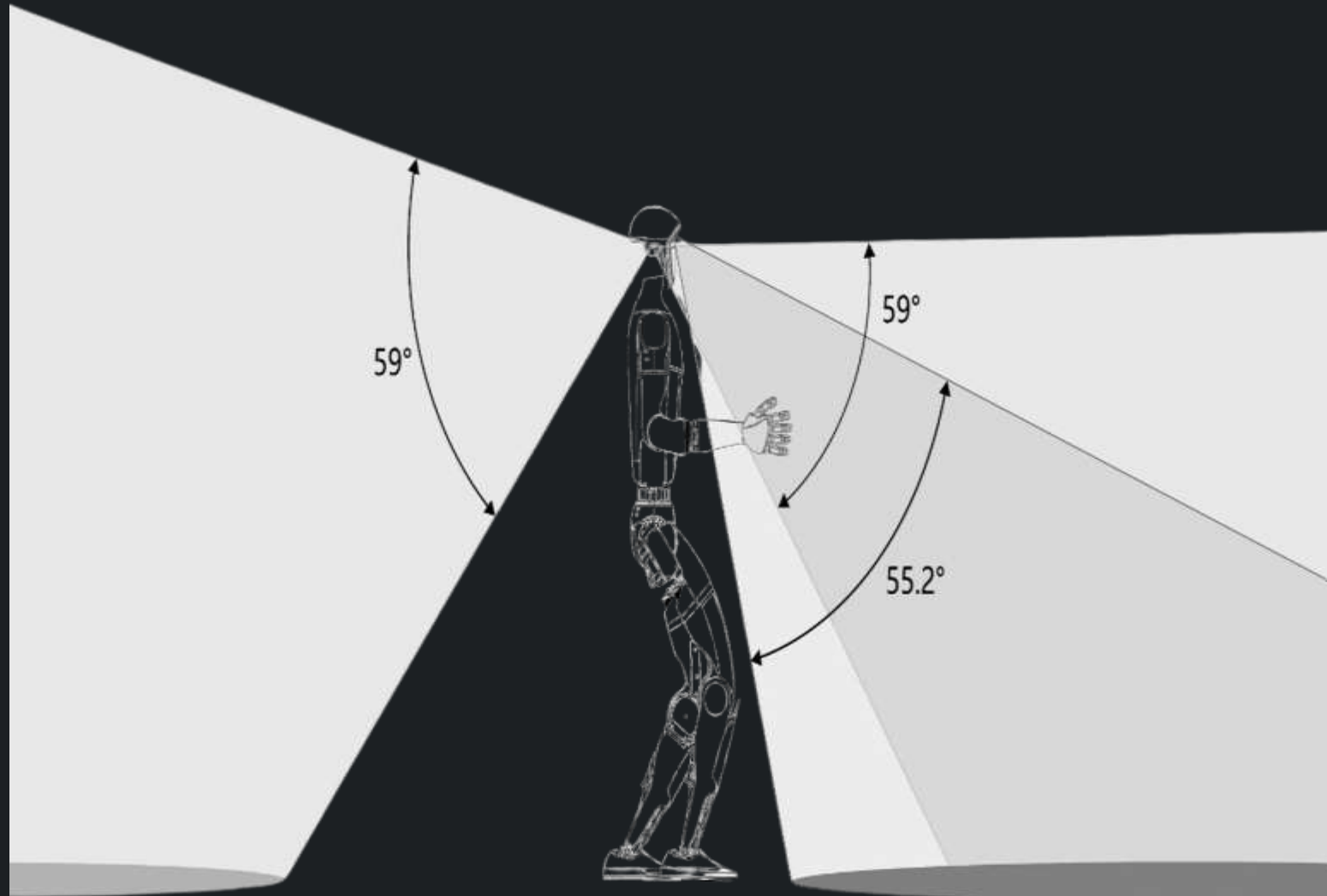
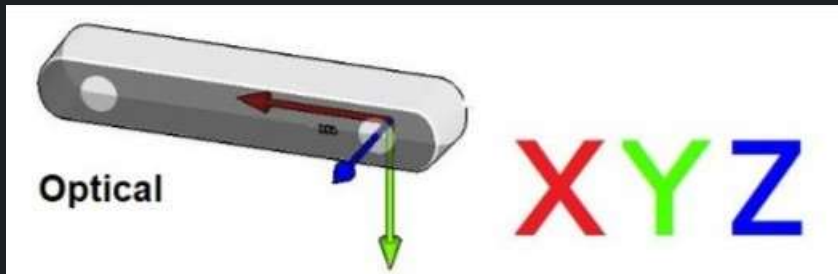


Überblick Unitree G1 Edu Motoren

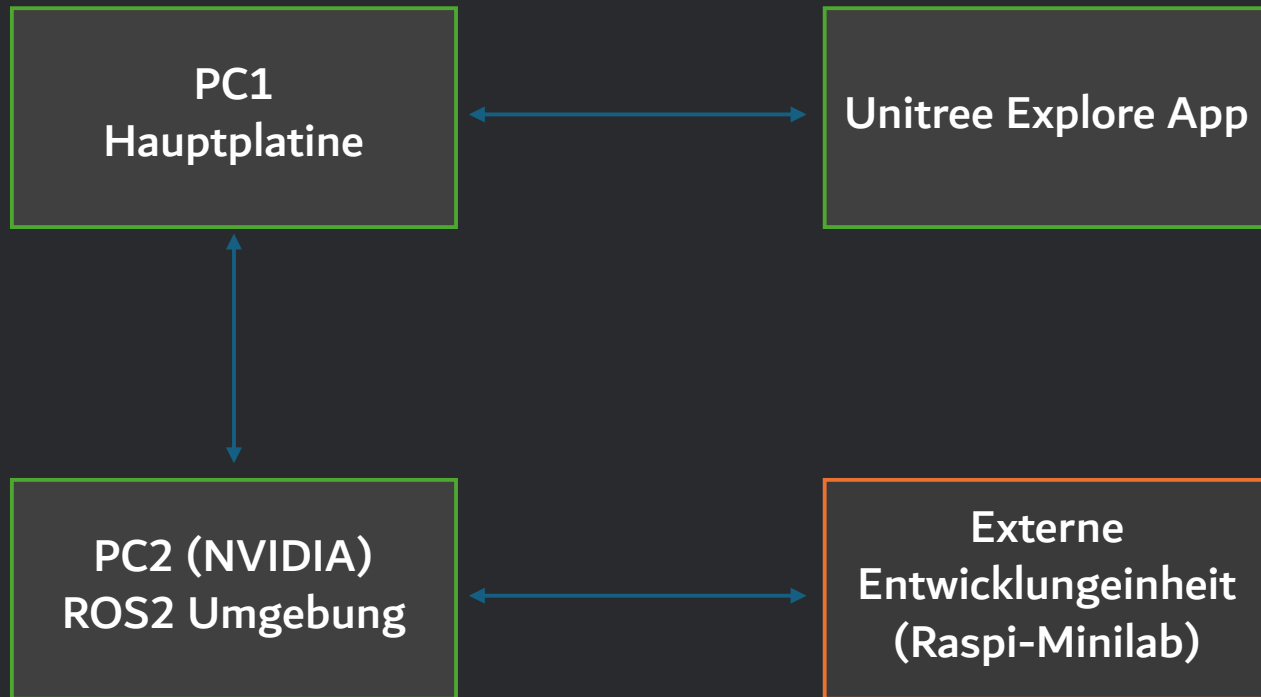


Überblick Unitree G1 Edu

- Sensoren
- 3D Lidar (Livox)
- RGB-D Kamera (Realsense)
- IMU



Überblick Unitree G1 Edu Hardware & Software



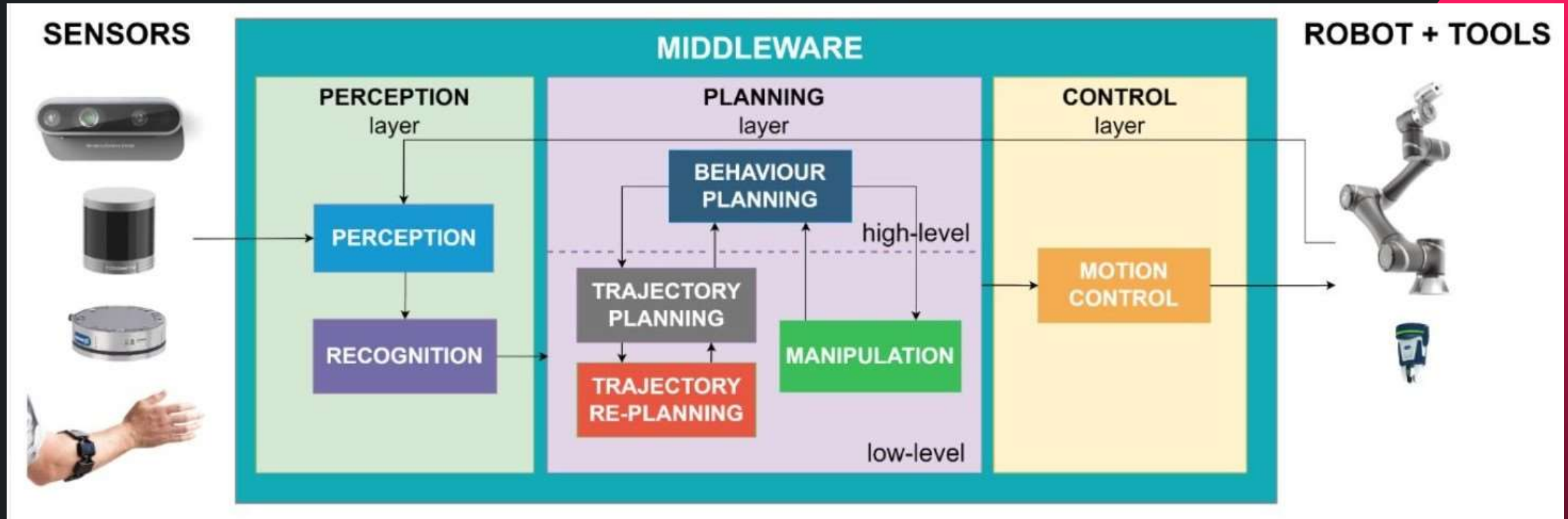
Remote Steuerung

- Bluetooth Verbindung
- AP Mode
- Wifi Mode



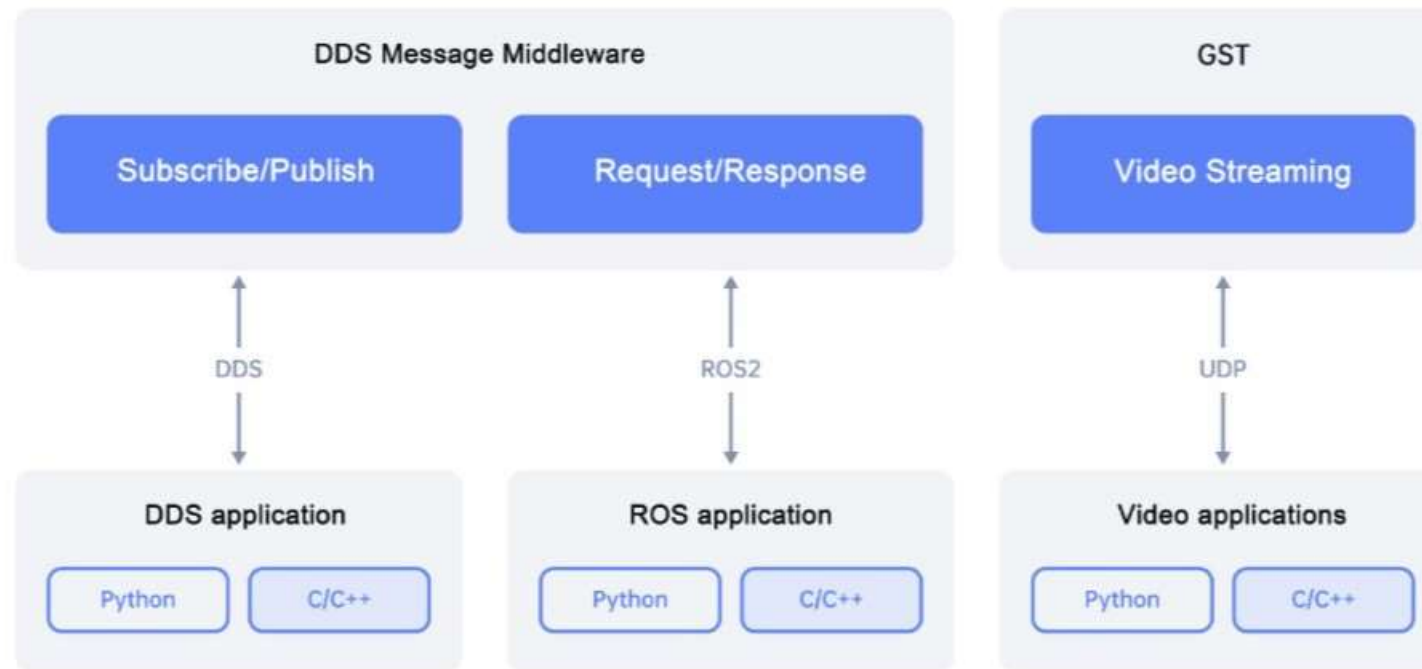
G1 Programmierung

ROS2 – High Level vs Low Level Programmierung



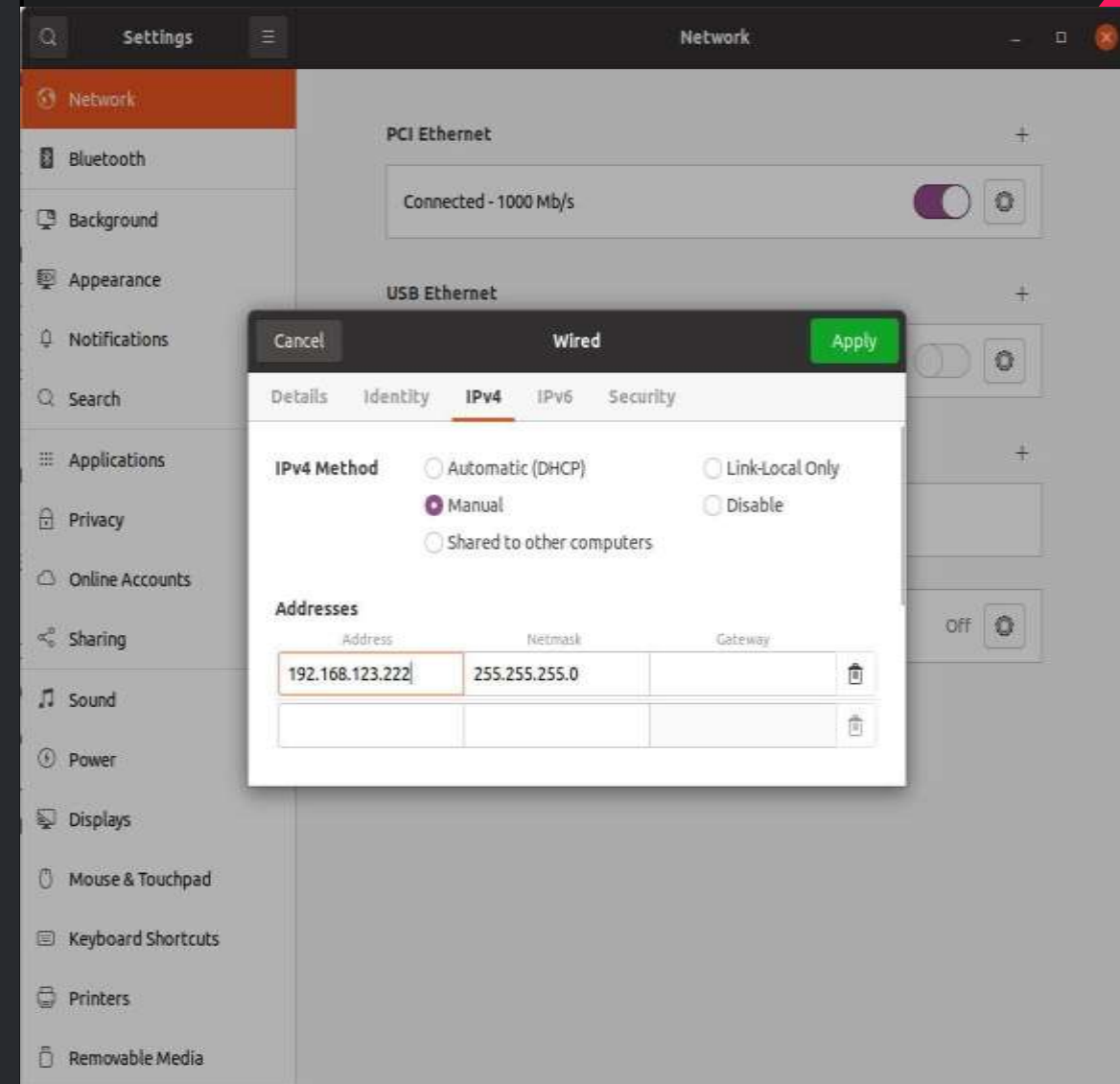
G1 Programmierung

SDK Overview



Verbindungsmethoden

- Ethernet / AP
- IP Konfiguration
- Bei IPv4 Einstellungen für die Schnittstelle, manuelles DHCP wählen
- Adresse 192.168.123.222
- Mask 255.255.255.0



Verbindungsmethoden

Interface name

```
liang@xiaoliangstd:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:e3:51:19:87 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.4.135 netmask 255.255.254.0 broadcast 192.168.5.255
    inet6 fe80::ee85:8ac4:f415:12f1 prefixlen 64 scopeid 0x20<link>
    ether 04:42:1a:96:33:79 txqueuelen 1000 (Ethernet)
    RX packets 1896133 bytes 1518864737 (1.5 GB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1164643 bytes 199679779 (199.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 16 memory 0xa1200000-a1220000

enxf8e43b808e06: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.123.222 netmask 255.255.255.0 broadcast 192.168.123.255
    inet6 fe80::2320:9805:e699:dd24 prefixlen 64 scopeid 0x20<link>
    ether f8:e4:3b:80:8e:06 txqueuelen 1000 (Ethernet)
    RX packets 8153 bytes 8827164 (8.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8384 bytes 2906296 (2.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```


Entwicklungsumgebung

Updates & Basic Tools

1) Update & upgrade system

bash

 Copy code

```
sudo apt update
```

```
sudo apt upgrade -y
```

2) Install base packages

bash

 Copy code

```
sudo apt install -y net-tools tree cmake git openssh-client openssh-server
```

Entwicklungsumgebung

Python 3.10

3) Install Python 3.10

Ubuntu doesn't always include 3.10 by default — add the Deadsnakes PPA and install: [Stack Overflow](#)

bash

 Copy code

```
sudo apt install -y software-properties-common
sudo add-apt-repository -y ppa:deadsnakes/ppa
sudo apt update
sudo apt install -y python3.10 python3.10-venv python3.10-dev
```

Optional: set it as the `python3` default (won't replace system Python):

bash

 Copy code

```
sudo update-alternatives --install /usr/bin/python3 python3 /usr/bin/python3.10 1
```

Entwicklungsumgebung

UV (Empfohlen)

Install uv

bash

 Copy code

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

bash

 Copy code

```
uv venv academy --python 3.10
```

Activate it:

bash

 Copy code

```
source academy/bin/activate
```

Entwicklungsumgebung

Abhängigkeiten Unitree_sdk2_python

```
cd ~  
sudo apt install python3-pip  
git clone https://github.com/unitreerobotics/unitree_sdk2_python.git  
cd unitree_sdk2_python  
pip3 install -e .
```



Entwicklungsumgebung

Abhängigkeiten

CycloneDDS

1. Error when `pip3 install -e .:`

```
Could not locate cyclonedds. Try to set CYCLONEDDS_HOME or CMAKE_PREFIX
```

This error mentions that the cyclonedds path could not be found. First compile and install cyclonedds:

```
cd ~
git clone https://github.com/eclipse-cyclonedds/cyclonedds -b releases/
cd cyclonedds && mkdir build install && cd build
cmake .. -DCMAKE_INSTALL_PREFIX=../install
cmake --build . --target install
```

Entwicklungsumgebung

Npm & codex (AI CLI) (OpenAI, Bezahlt)

Install

bash

 Copy code

```
sudo npm install -g @openai/codex
```

Update

bash

 Copy code

```
sudo npm update -g @openai/codex
```

Entwicklungsumgebung

Alternativ: Gemini-CLI (Google, Kostenlos)

Optional Tools

Gemini CLI (npm)

Official package name is typically `@google/gemini-cli`: Milvus

bash

 Copy code

```
sudo npm install -g @google/gemini-cli
```

To update later:

bash

 Copy code

```
sudo npm update -g @google/gemini-cli
```

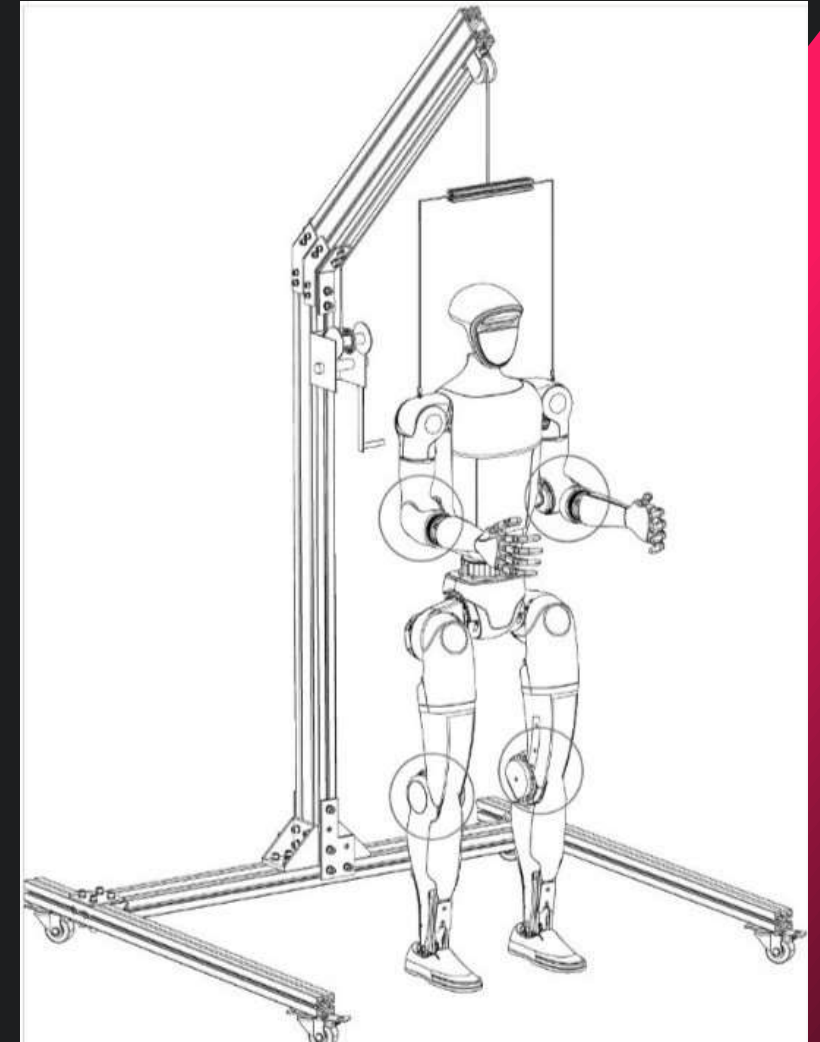

G1 Programmierung

- Sicherheitsanweisungen:

https://support.unitree.com/home/en/G1_developer/quick_start

- Debugging:

https://support.unitree.com/home/en/G1_developer/common_istakes_and_definiti_ons



Entwicklungsumgebung

Akademie Repository

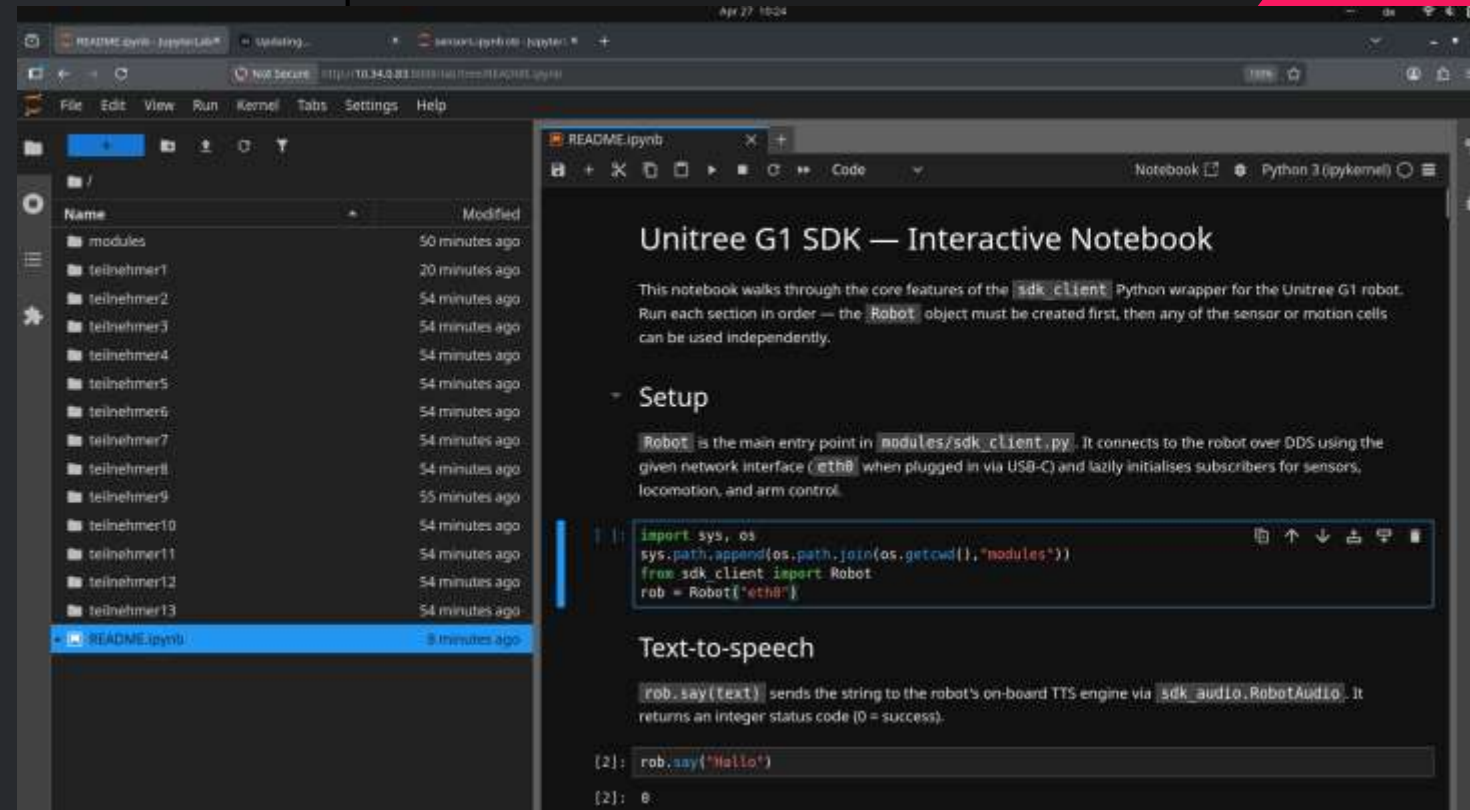


https://github.com/EF-Robotics-Humanoid-Academy/ef_ws

```
(ef_ws) unitree@ubuntu:~/ef_ws/repos/ef_ws/q1$ tree -L 2
.
├── dev
│   ├── dashboards
│   ├── edge_runtime
│   ├── gi_sport_node_ros
│   └── old_scripts
├── docs
│   ├── fsm_cheatsheet.html
│   ├── geoff_explorer.md
│   ├── geoff_stack.md
│   ├── geoff_stack.png
│   ├── how_it_works.md
│   ├── index.md
│   ├── lidar_cheatsheet.html
│   ├── lidar_docs.html
│   ├── notes.txt
│   ├── quick_start.md
│   ├── realsense_streaming_docs.html
│   ├── requirements.txt
│   ├── scripts.md
│   └── troubleshooting.md
├── modules
│   ├── notebooks
│   │   └── pycache
│   ├── scripts
│   ├── sdk_audio.py
│   ├── sdk_boot.py
│   ├── sdk_client.py
│   ├── sdk_hand.py
│   ├── sdk_sensors.py
│   ├── sdk_slam.py
│   ├── secure_boot.py
│   └── Untitled.ipynb
├── REQUIREMENTS.txt
├── ros_envs
│   ├── foxy
│   └── noetic
├── scripts
│   ├── basic
│   ├── dash_robot_control.log
│   ├── dev
│   ├── navigation
│   └── pycache
└── sim
    ├── G1_rviz_simulation-main
    ├── lafani_retargeting_dataset
    ├── mujoco
    ├── pybullet
    └── unitree_robots
```

Jupyter interactive Notebook:

- <http://10.34.0.206:8888>



Aufgabe 1

- Neues Jupyter notebook öffnen
- `from sdk_client import Robot`
- `rob = Robot(„eth0“)`
- `rob.say(„...“)`



AI CLI Einführung

- codex / claude
- Benutzung vom resume option
- Prompt engineering (Persona, context, beispiele)
- Sub agents custom



Aufgabe 2

- codex in jupyter lad öffnen
- codex über sdk modules lernen lassen
- Ein skript mit codex erstellen um ein custom mode zu erstellen
 - Modus Name aussprechen
 - Lichtfarbe mit einem python Thread konstant zu halten (persistent_headlight_control.py als context für codex geben)

